



Oral Cancer

DR. WAJEEHA SHAMS ANSARI

Learning Objectives

By the end of this presentation, students should be able to:

1. Define oral cancer.
2. Identify major risk factors.
3. Explain the role of DNA mutations and oxidative stress in oral carcinogenesis.
4. Describe the significance of the p53 tumor suppressor gene.
5. Recognize common premalignant lesions.
6. Discuss basic methods of diagnosis and prevention.

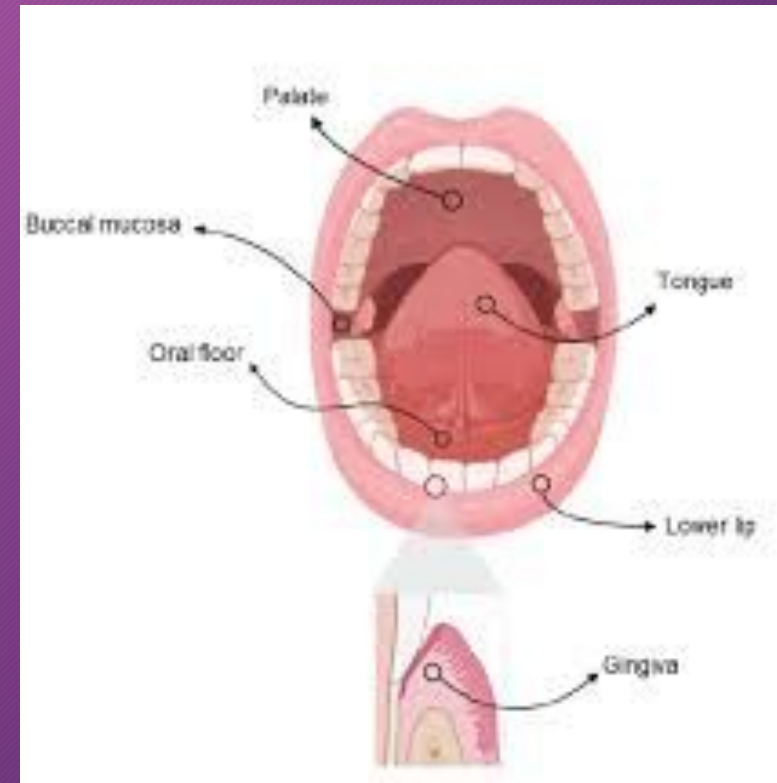
Introduction

Definition

- Oral cancer is a malignant tumor arising from the tissues of the oral cavity.
- **Most Common Type**
- **Oral Squamous Cell Carcinoma (OSCC) (90-95% of cases)**

Common Sites

- Tongue
- Buccal mucosa
- Floor of mouth
- Lips
- Gingiva



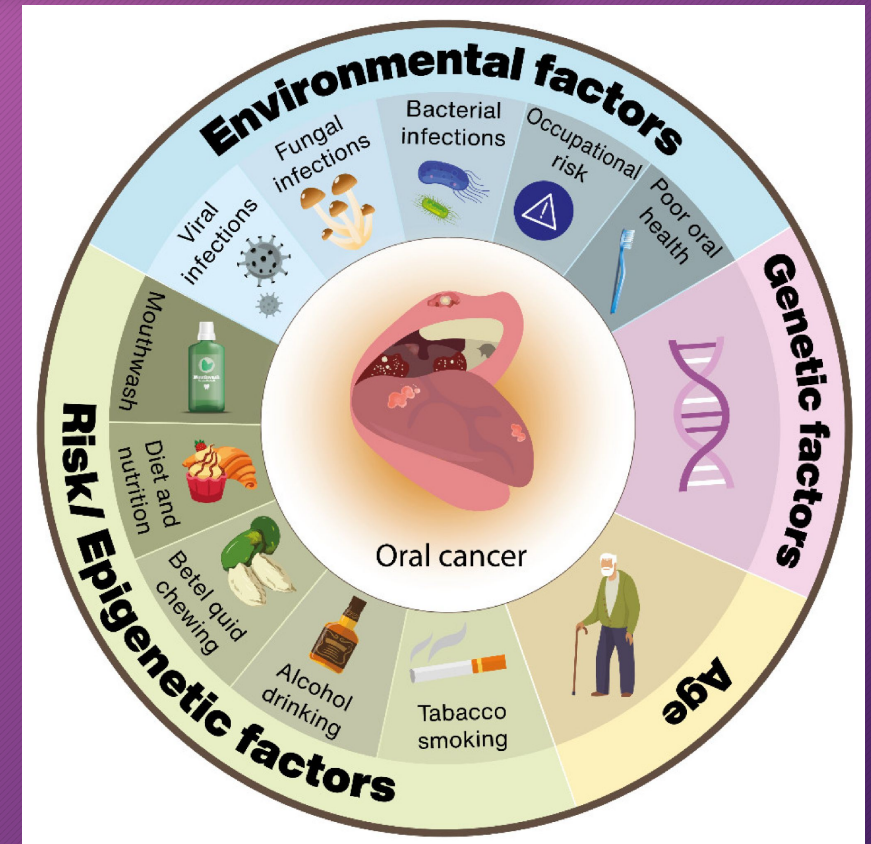
Risk Factors

Major Risk Factors

- Tobacco smoking
- Smokeless tobacco (gutka, naswar, paan)
- Areca (betel) nut chewing
- Alcohol consumption
- HPV infection
- Prolonged UV radiation exposure
- Chronic inflammation or scars
- Immunosuppression

Importance

- Most cases are associated with preventable lifestyle factors.



Biochemical Basis of Oral Cancer

Cancer develops due to:

- DNA damage
- Genetic mutations
- Abnormal cell growth
- Failure of normal cell death (apoptosis)

Outcome

- Normal cell → Mutation → Uncontrolled proliferation → Cancer

Role of Reactive Oxygen Species (ROS)

ROS Include:

- Superoxide radicals
- Hydroxyl radicals
- Hydrogen peroxide

Effects

- DNA damage
- Protein damage
- Lipid peroxidation

These changes increase the risk of malignant transformation.

p53 Gene - "Guardian of the Genome"

Functions

- Repairs damaged DNA
- Stops cell cycle when DNA is damaged
- Initiates apoptosis

In Oral Cancer

Mutation of p53 leads to:

- Survival of damaged cells
- Accumulation of mutations
- Tumor formation

Premalignant Lesions

Leukoplakia

- White patch that cannot be scraped off

Erythroplakia

- Red patch with high malignant potential

Oral Submucous Fibrosis

- Common in areca nut users



Early detection can prevent cancer development.

Clinical Features

Common Signs and Symptoms

- Non-healing oral ulcer
- White or red patches
- Pain or burning sensation
- Difficulty chewing or swallowing
- Enlarged cervical lymph nodes
- Persistent scaly or crusted lesion
- Firm red nodule or plaque
- Pain, bleeding, or rapid growth in advanced cases

Diagnosis and Prevention

Diagnosis

- Clinical examination
- Biopsy (gold standard) with histopathological examination
- Imaging for staging when indicated

Prevention

- Avoid tobacco and areca nut
- Limit alcohol intake
- Maintain oral hygiene
- Regular dental check-ups

Histopathology

- Invasive nests of atypical squamous cells
- Keratin pearls
- Intercellular bridges
- Cellular pleomorphism and mitotic activity

Treatment

- Surgical excision (gold standard)
- Mohs micrographic surgery (selected cases)
- Radiation therapy
- Chemotherapy/immunotherapy for advanced disease

Prognosis

- Generally favorable when detected early
- Risk of local invasion and metastasis if untreated

Key Message: Early detection and complete surgical removal significantly improve outcomes.

Conclusion

- Oral cancer is a common and preventable malignancy.
- Tobacco and areca nut are major risk factors.
- DNA damage, ROS, and p53 mutations are important biochemical factors.
- Early detection and prevention greatly improve prognosis.

thank you!

